

Introduction to Many-Body Quantum Information

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Created: March 4, 2026

Updated: May 9, 2026

I Quantum gates and circuits

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Main references for this section:

- Michael A. Nielsen, & Isaac L. Chuang. *Quantum Computation and Quantum Information*.
- Giuliano Benenti, Giulio Casati, & Giuliano Strini. *Principles of Quantum Computation and Information*.

1.1 Concept of quantum computer

1.1.1 Motivation

In the Schrödinger picture, the evolution of a closed quantum system is described by

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad (1)$$

where $U(t)$ is a unitary operator satisfying $U^\dagger(t)U(t) = I$. For a time-independent Hamiltonian H , the unitary evolution operator takes the form

$$U(t) = e^{-iHt}. \quad (2)$$

Notice that exactly simulating a quantum many-body system on classical computers generally requires exponentially large computational resources, due to the exponential growth of the Hilbert-space dimension with system size.

A basic idea to overcome this difficulty is the following.

- Suppose we have a controllable quantum system Q_0 , meaning that we can manipulate its state $|\psi\rangle$ through a set of physically implementable operations so that it evolves into a desired state $|\psi'\rangle$. Then we may investigate another quantum system Q of interest by mimicking or simulating the behavior of Q using Q_0 .

The controllable system Q_0 is exactly what we call a *quantum computer*. This naturally leads to two fundamental questions:

1. How can we physically build such a controllable quantum system (the *hardware* problem)?
2. Given a set of available unitary operations, which transformations U should be implemented in order to evolve $|\psi\rangle$ into a desired state $|\psi'\rangle$ (the *software* problem)? This requires the mathematical design of suitable unitary transformations U . Such procedures are known as *quantum algorithms*.

1.1.2 From classical computation to quantum computation

Since the theory of classical computation is well established, it is natural to develop the theory of quantum computation by drawing analogies with it.

In classical computing, the elementary unit of information is a *bit*, which is a binary variable

$$c \in \{0, 1\}, \quad (3)$$

physically realized by logic circuits implemented on semiconductor chips.

Quantum mechanics, however, permits the superposition of states. This motivates the definition of the *quantum bit*, or *qubit*, whose general state takes the form

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad (4)$$

where $\alpha, \beta \in \mathbb{C}$ satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$.

A classical bit can therefore be viewed as a special case of a qubit, corresponding to the computational-basis states $|0\rangle$ and $|1\rangle$, i.e.,

$$(\alpha, \beta) = (1, 0) \quad \text{or} \quad (0, 1). \quad (5)$$

Physically, a qubit can be realized by a quantum two-level system, such as the spin states of an electron, two energy levels of an atom, or the polarization states of a photon.

Remark 1.1. Although a qubit is described by two complex amplitudes, the amount of classical information obtained from measuring a single qubit is still only one bit. The advantage of quantum computation arises from coherent manipulation and interference among amplitudes in multi-qubit systems, instead of simple superposition.

1.2 Single qubit

1.2.1 Bloch sphere and single-qubit rotation gates

Based on the definition above, a general qubit state $|q\rangle$ can be written as

$$|q\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle, \quad (6)$$

where the computational-basis states are

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (7)$$

Thus the qubit can be represented as

$$|q\rangle = \begin{bmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{bmatrix}. \quad (8)$$

Since an overall global phase of a quantum state is physically irrelevant, every pure qubit state can be uniquely represented by a point on a unit sphere known as the *Bloch sphere* (see Fig. 1). The corresponding Bloch vector is

$$\vec{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta). \quad (9)$$

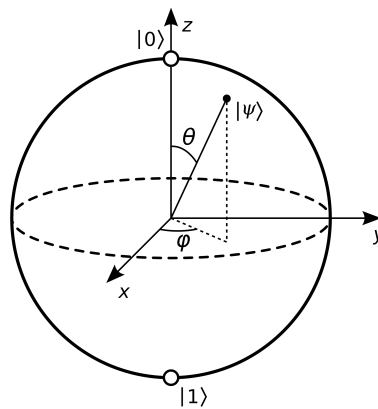


Figure 1: Bloch sphere (image from Wikipedia).

Under a unitary transformation $U \in SU(2)$, the qubit state $|q\rangle \rightarrow U|q\rangle$. On the Bloch sphere this induces a rotation of the Bloch vector

$$\vec{r} \longrightarrow \vec{r}'. \quad (10)$$

Thus single-qubit unitary operations correspond geometrically to rotations of the Bloch sphere.

Remark 1.2. The Bloch sphere represents pure qubit states up to a global phase. Single-qubit unitary gates belong to $SU(2)$ and act as rotations of the Bloch vector in $SO(3)$. The mapping $SU(2) \rightarrow SO(3)$ is two-to-one, since U and $-U$ produce the same rotation of the Bloch sphere.

A convenient set of generators of $SU(2)$ is

$$J^j = \frac{\sigma^j}{2}, \quad (11)$$

where σ^j are the *Pauli operators*.

Remark 1.3. In quantum computation, Pauli operators are usually denoted as

$$X \equiv \sigma^x, \quad Y \equiv \sigma^y, \quad Z \equiv \sigma^z. \quad (12)$$

Using these operators, we define the single-qubit *rotation gates*

$$R_x(\theta) = e^{-i\theta X/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} X = \begin{bmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix}, \quad (13)$$

$$R_y(\theta) = e^{-i\theta Y/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} Y = \begin{bmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix}, \quad (14)$$

$$R_z(\theta) = e^{-i\theta Z/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} Z = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}. \quad (15)$$

These rotations constitute fundamental single-qubit operations. In fact, any single-qubit unitary operator $U \in SU(2)$ can be decomposed into a sequence of such rotations.

Since a global phase does not affect a quantum state, two gates that differ only by an overall phase are usually regarded as equivalent. For example, one also often writes

$$R_z(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}. \quad (16)$$

1.2.2 Frequently-used single-qubit gates

For convenience, several commonly used single-qubit gates are predefined.

Pauli gates.

$$X \equiv iR_x(\pi) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (17)$$

$$Y \equiv iR_y(\pi) = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad (18)$$

$$Z \equiv iR_z(\pi) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (19)$$

- X is also known as the *NOT gate*, since it flips the computational-basis states:

$$X |0\rangle = |1\rangle, \quad X |1\rangle = |0\rangle. \quad (20)$$

Hadamard gate.

$$H \equiv iR_x(\pi)R_y\left(\frac{\pi}{2}\right) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (21)$$

- It satisfies the identity

$$H = \frac{X + Z}{\sqrt{2}}. \quad (22)$$

- The Hadamard gate maps computational-basis states to equal superpositions:

$$H |0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H |1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Thus it plays a central role in generating quantum superposition.

Phase gate or S gate.

$$S := R_z\left(\frac{\pi}{2}\right) = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \quad (23)$$

T gate.

$$T := R_z\left(\frac{\pi}{4}\right) = e^{i\frac{\pi}{8}} \begin{bmatrix} e^{-i\frac{\pi}{8}} & 0 \\ 0 & e^{i\frac{\pi}{8}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix} \quad (24)$$

- The T gate is particularly important in fault-tolerant quantum computation. It introduces *non-stabilizer resources*, often referred to as *magic*, which are necessary for universal quantum computation within the stabilizer framework.

1.3 Multiple qubits

1.3.1 Universal quantum computation

Suppose $|\psi\rangle$ and $|\psi'\rangle$ are two arbitrary states of an N -qubit quantum system. If there exists a sequence of quantum gates that can transform $|\psi\rangle$ into a state arbitrarily close to $|\psi'\rangle$, then such gates provide universal control of the system.

More generally, a set of quantum gates is called a *universal set of quantum gates* if any unitary operator U acting on N qubits can be approximated to arbitrary precision by a finite sequence of gates from this set.

A quantum computer that possesses such a universal gate set is called a *universal quantum computer*. In contrast, devices that only implement a restricted class of operations are referred to as *non-universal* or *specialized quantum devices*. An example is a *quantum annealer*, which is designed to solve optimization problems via adiabatic quantum evolution but does not support universal quantum computation.

Remark 1.4. Apparently, for single-qubit systems, the collection of all single-qubit unitary gates, i.e. the group $SU(2)$, already forms a universal set of operations.

However, when considering a multi-qubit system, single-qubit unitaries alone are not sufficient for universal quantum computation, since they cannot generate entanglement between qubits.

For example, the *Bell state*

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (25)$$

cannot be prepared using single-qubit gates starting from $|00\rangle$.

To generate entanglement, we introduce the *controlled-NOT (CNOT) gate*, which is a two-qubit unitary operation. One qubit is designated as the *control qubit*, and the other as the *target qubit*. The action of the CNOT gate is

- If the control qubit is in the state $|1\rangle$, apply an X gate to the target qubit.
- Otherwise, do nothing.

The matrix representation of the CNOT gate (in the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$) is

$$\text{CNOT}_{j \rightarrow k} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \text{CNOT}_{k \rightarrow j} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad (26)$$

where the subscripts indicate the control and target qubits. We assume the ordering that qubit j is written to the left of qubit k in the tensor-product basis, i.e.

$$|ab\rangle = |a\rangle_j \otimes |b\rangle_k. \quad (27)$$

Theorem 1.5. *Arbitrary single-qubit rotation gates together with the CNOT gate form a universal set of quantum gates for any N -qubit system.*

This universal set of quantum gates is not unique. For example, the finite gate set

$$\{H, S, T, \text{CNOT}\} \quad (28)$$

also forms a universal set of quantum gates.

1.3.2 Diagrams of quantum circuits

In this section we introduce the graphical representation of quantum circuits.

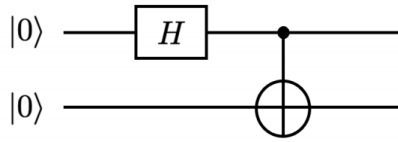


Figure 2: A simple example of a quantum circuit. The square labeled H denotes a Hadamard gate. The two-qubit gate shown here is a CNOT gate: the solid dot represents the control qubit, while the circle with a cross represents the target qubit.

A simple example is shown in Fig. 2. In such diagrams,

- Each horizontal wire corresponds to a qubit;
- The $|0\rangle$ on the left side of a wire denotes the initial state of that qubit;
- Gates act on the qubits sequentially from left to right, representing the order of operations in time.

The circuit in Fig. 2 prepares the Bell state

$$|00\rangle \longrightarrow \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad (29)$$

as discussed previously.

1.3.3 Frequently used multi-qubit gates

SWAP gate. The SWAP gate exchanges the states of two qubits:

$$\text{SWAP} : |a\rangle |b\rangle \longrightarrow |b\rangle |a\rangle. \quad (30)$$

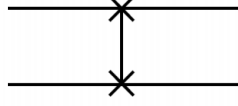


Figure 3: Circuit representation of the SWAP gate on two qubits.

For two qubits, its matrix representation is

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (31)$$

The graphical notation of the SWAP gate is shown in Fig. 3.

Multi-controlled- U gate ($C^m(U)$). A multi-controlled- U gate is a generalization of the CNOT gate. It consists of m control qubits and a unitary operation U acting on n target qubits.

- The operation U is applied to the target qubits if and only if all control qubits are in the state $|1\rangle$; otherwise the target qubits remain unchanged.
- The CNOT gate is therefore also called the CX gate.
- The C^2X gate (double-controlled NOT) is known as the *Toffoli gate*.
- The controlled-SWAP gate ($C(\text{SWAP})$) is known as the *Fredkin gate*.

1.4 The Deutsch–Jozsa algorithm

1.4.1 The problem

The Deutsch–Jozsa algorithm is one of the earliest examples showing that quantum computers can outperform classical computers for certain oracle problems.

Suppose we are given a Boolean function

$$f : \{0, 1\}^N \rightarrow \{0, 1\}. \quad (32)$$

We are promised that the function is either

- *constant*: $f(x)$ takes the same value (0 or 1) for all x ;
- *balanced*: $f(x) = 0$ for exactly half of the inputs and $f(x) = 1$ for the other half.

The goal is to determine whether f is constant or balanced.

Remark 1.6. In the classical deterministic case, one must evaluate $f(x)$ for $(2^{N-1} + 1)$ different inputs in the worst case to guarantee the correct answer.

1.4.2 The algorithm

We assume access to a quantum oracle U_f defined by

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle, \quad (33)$$

where $\oplus$ denotes addition modulo 2.

The full algorithm is illustrated in Fig. 4.

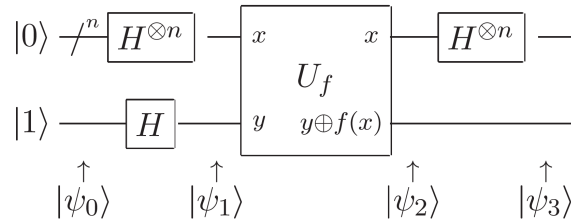


Figure 4: The Deutsch-Jozsa algorithm (figure from Nielsen's book.)

The initial state is prepared as

$$|\psi_0\rangle = |0\rangle^{\otimes N} \otimes |1\rangle. \quad (34)$$

After applying Hadamard gates to all qubits, the state becomes

$$|\psi_1\rangle = \sum_x \frac{|x\rangle}{\sqrt{2^N}} \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (35)$$

Next, we apply the oracle U_f , which gives

$$|\psi_2\rangle = \sum_x \frac{(-1)^{f(x)} |x\rangle}{\sqrt{2^N}} \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (36)$$

To proceed, note that for a bit string $x = x_{N-1}x_{N-2} \cdots x_0$, applying a Hadamard gate to the j th qubit yields

$$H |x_j\rangle = \frac{1}{\sqrt{2}} \sum_{z_j \in \{0,1\}} (-1)^{x_j z_j} |z_j\rangle. \quad (37)$$

Therefore,

$$\begin{aligned} H^{\otimes N} |x\rangle &= \frac{\sum_{z_{N-1}, z_{N-2}, \dots, z_0} (-1)^{x_{N-1}z_{N-1} + x_{N-2}z_{N-2} + \dots + x_0 z_0} |z\rangle}{\sqrt{2^N}} \\ &\equiv \frac{1}{\sqrt{2^N}} \sum_{\mathbf{z}} (-1)^{\mathbf{x} \cdot \mathbf{z}} |\mathbf{z}\rangle, \end{aligned} \quad (38)$$

where we define

$$\mathbf{x} = (x_{N-1}, x_{N-2}, \dots, x_0), \quad \mathbf{z} = (z_{N-1}, z_{N-2}, \dots, z_0). \quad (39)$$

Applying $H^{\otimes N}$ again to the first N qubits of $|\psi_2\rangle$ gives

$$|\psi_3\rangle = \sum_{\mathbf{z}} \sum_x \frac{(-1)^{\mathbf{x} \cdot \mathbf{z} + f(x)} |z\rangle}{2^N} \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (40)$$

Finally, we measure the first N qubits of $|\psi_3\rangle$. The probability of obtaining $|0\rangle^{\otimes N}$ is

$$P(|z\rangle = |0\rangle^{\otimes N}) = \left| \sum_x \frac{(-1)^{f(x)}}{2^N} \right|^2. \quad (41)$$

- If f is constant, the probability of obtaining $|0\rangle^{\otimes N}$ is 1;
- If f is balanced, the probability of obtaining $|0\rangle^{\otimes N}$ is 0.

Therefore, a single evaluation of the oracle determines with certainty whether f is constant or balanced, demonstrating an exponential speedup over classical deterministic algorithms.